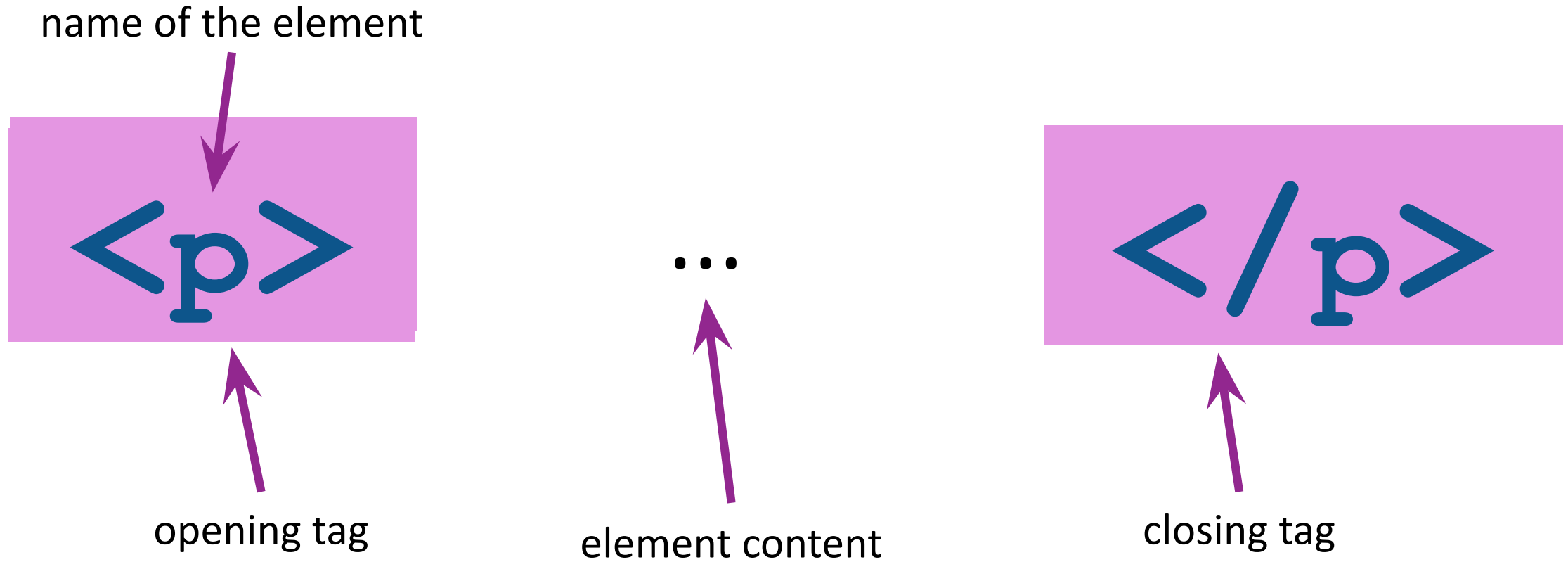


Lecture 2

- HTML element - syntax
- HTML element categories
- head
- Structuring the body
- Global/non-global attributes
 - Strings in HTML
- Semantic tags (HTML5)
 - suggested page structure
 - semantic vs. generic block/inline
- HTML Content model (HTML5)
 - Links as an example of flow
- Character encoding with UTF-8
- Character escapes

HTML element - syntax



HTML element – syntax (2)

no space allowed

space required

no spaces allowed

```
<p class="info"> . . </p>
```

attribute

attribute name

attribute value

HTML element categories

- Main root: `<html>` and inside it: `<head>` `<body>`
- Document metadata in head:
`<title>` `<base>` `<link>` `<meta>` `<style>`
- Section elements – structuring content in body:
`<section>` `<nav>` `<article>` `<aside>` `<h1>.. <h6>` `<hgroup>` `<header>` `<footer>` `<address>`
- Other grouping elements for content:
`<p>` `<hr>` `<br>` `<pre>` `<blockquote>` `<ol>` `<ul>` `<li>` `<dl>` `<dt>` `<dd>` `<figure>` `<figcaption>` `<div>`
- Text level elements – styling text segments:
`<a>` `<em>` `<strong>` `<small>` `<s>` `<cite>` `<q>` `<dfn>` `<abbr>` `<time>` `<code>` `<var>` `<samp>` `<kbd>` `<sub>`
`<sup>` `<i>` `<b>` `<u>` `<mark>` `<ruby>` `<rt>` `<rp>` `<bdi>` `<bdo>` `<span>`
- Embedded content:
`<img>` `<iframe>` `<embed>` `<object>` `<param>` `<video>` `<audio>` `<source>` `<track>` `<canvas>` `<map>` `<area>`
- Tables:
`<table>` `<caption>` `<colgroup>` `<col>` `<tbody>` `<thead>` `<tfoot>` `<tr>` `<td>` `<th>`
- Forms:
`<form>` `<fieldset>` `<legend>` `<label>` `<input>` `<button>` `<select>` `<datalist>`
`<optgroup>` `<option>` `<textarea>` `<keygen>` `<output>` `<progress>` `<meter>`

HTML
elementsHTML
elements

head section of document

- The first child of a root: `<html>`
- Possible elements:
 - `<title>` title on the page's tab
 - `<meta>` wide variety of different information about the page (charset, viewport size, redirecting, etc.)
 - `<link>` links the document to external resources
 - `<style>` contains style rules (CSS –rules) for the page
 - `<base>` default path for the links given in the documents

```
<!DOCTYPE html>
<html lang="fi">
  <head>
    <meta charset="UTF-8">
    <title>Title on the tab</title>
    <link res="stylesheet"
          href="styles.css">
  </head>
  <body>
    <!-- the content of the page -->
  </body>
</html>
```



`<head>`

body section of document

```
<!DOCTYPE html>
<html lang="fi">
<head>
  <meta charset="UTF-8">
  <title>Title on the tab</title>
  <link res="stylesheet"
        href="styles.css">
</head>
<body>
  <!-- the content of the page -->
</body>
</html>
```

- The content of the document is written into the body
- In the `<body>`
 - 1) define the structure of the document
 - 2) mark-up the semantic parts of the text

Global attributes

- Global attributes are attributes used in all elements, such as:

`accesskey, class, contenteditable, contextmenu, dir, draggable, dropzone, hidden, id, lang, spellcheck, style, tabindex, title, translate`

- The most important attributes are id and class
- In addition, multiple unstandardized ones



Global attributes



Global attributes

id attribute

- `id` should identify an element uniquely within a document
 - the element with `id part1` is referred by using #-character:
`#part1`

```
<h1 id="part1">Introduction to the topic</h1>
```

- the `id` can be used, for example, to create a link to this position in the html-document, e.g., `index.html#part1`

class attribute

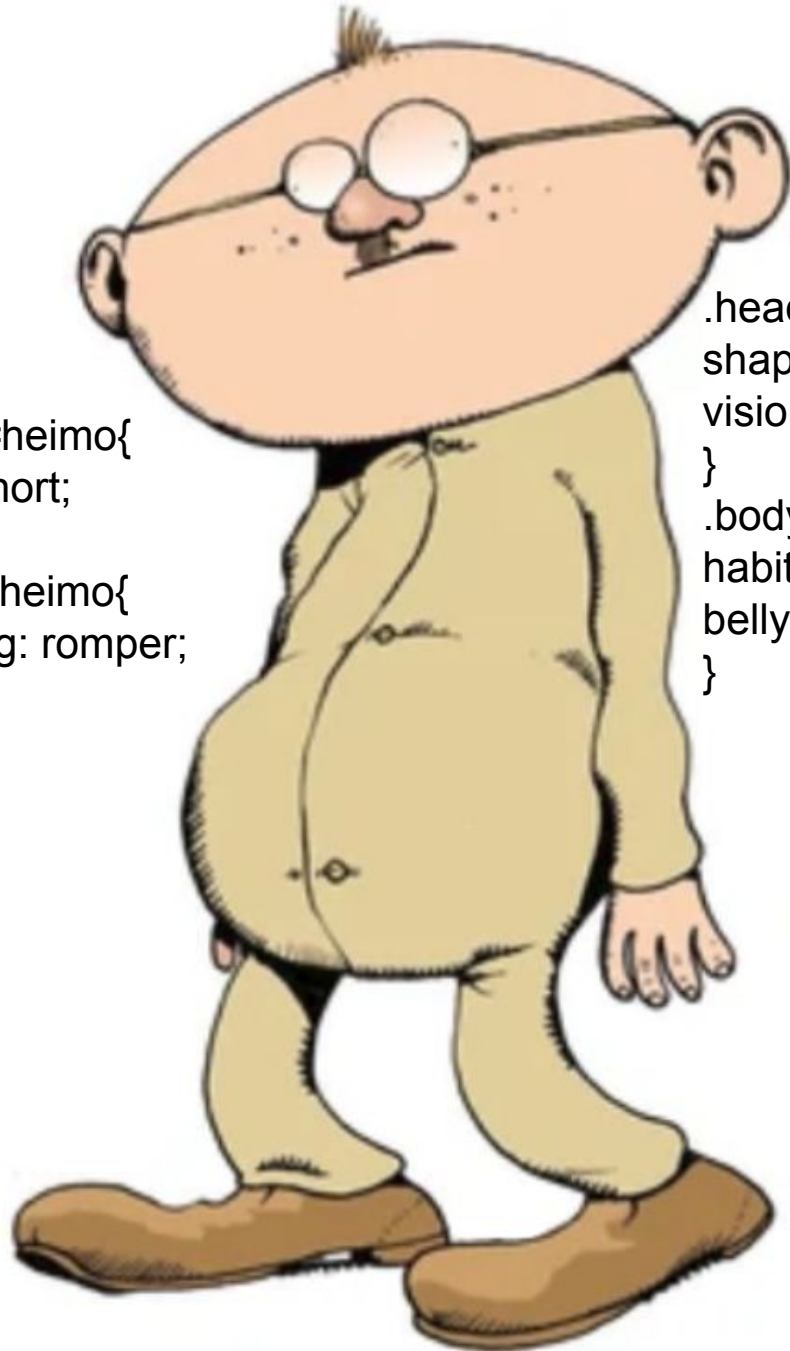
- class attribute defines that the element belongs to a certain class
- the elements in the class `ingress` are referred with using a dot:
`.ingress`

```
<p class="ingress">
```

```
  This chapter is marked to belong to  
  ingress type of paragraphs. In css  
  all ingresses can be laid out in  
  similar way.
```

```
</p>
```

```
.head#heimo{
hair: short;
}
.body#heimo{
clothing: romper;
}
```



```
.head.vesa{
shape: round;
vision: impaired;
}
.body.vesa{
habitus: floppy;
belly: abundant;
}
```



```
.head#niilo{
hair: ponytail;
}
.body#niilo{
clothing: jeans t-shirt;
}
```

In Fingerpori lives Heimo Vesa, a researcher;

In Fingerpoli works Heimo's cousin Niilo, a nerd

Non-global attributes

- The attributes control the behaviour and looks of some elements, e.g., lists:

```
<ol>  
  <li>aamiainen</li>  
  <li>lounas</li>  
  <li>päivällinen</li>  
</ol>
```



try out



```
<ol type="a">  
  <li>aamiainen</li>  
  <li>lounas</li>  
  <li>päivällinen</li>  
</ol>
```

- Reference table for all attributes



Attributes



Attributes

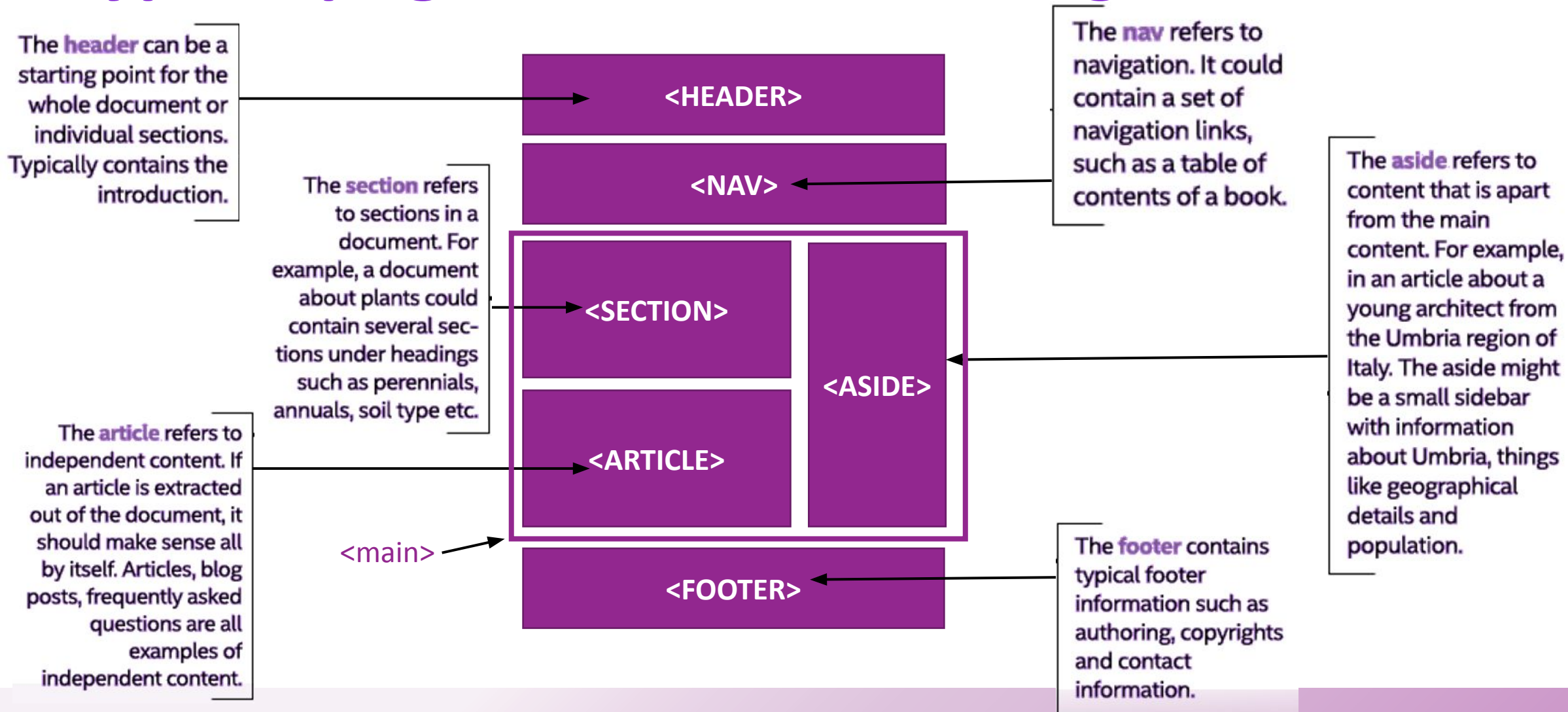
Strings in HTML

- Strings are defined with quotation marks (single (') or double ("))
 - attribute values, for instance, are given as strings, next styles are interchangeable
`<p id="part1">` `<p id='part1'>` `<p id=part1>`
 - because a browser knows to expect a string, quotation marks can be left out, which is not always the case, e.g. `<p class="emphasized noprint">`
in interpreting the attribute value, the quotation marks here are decisive.

Do use quotation marks (good practise)

New in HTML5: semantic tags

Typical page with semantic tags



Structuring the body semantically

- Structure is defined with the elements

 `<address>` contact information for a person or organization

 `<article>` self contained part which could be reused as a whole 

 `<aside>` portion only indirectly related to the main textual content

 `<footer>` typically includes information like, authors, copyrights, contacts

 `<header>` groups the introductory content of a section 

`<h1>..<h6>` six levels of section headers

 `<nav>` contains navigation links

`<section>` section with no more specific semantic element to represent it, typically it has a heading 

- How to use heading in sections?

 See  and 
Using `<section>`
element

 Content
sectioning

Semantic vs. generic block-level elements

- How to decide which one to use (block-level elements)?
 - does the element have a semantic role on the page structure

○ YES

Use the semantic block-level elements:

`article, aside,
blockquote, br, dl, dd,
dt, figcaption, figure,
footer, header, hr, li,
main, nav, ol, pre,
section, ul`

○ NO

Use the generic content grouping element

`div`

Semantic vs. generic inline elements

Inline elements with semantic meaning

`<a>`, `<em>`, `<strong>`, `<small>`,
`<s>`, `<cite>`, `<q>`, `<dfn>`, `<abbr>`,
`<time>`, `<code>`, `<var>`, `<samp>`,
`<kbd>`, `<sub>`, `<sup>`, `<i>`, `<b>`,
`<u>`, `<mark>`, `<ruby>`, `<rt>`, `<rp>`,
`<bdi>`, `<bdo>`

no predefined semantic meaning

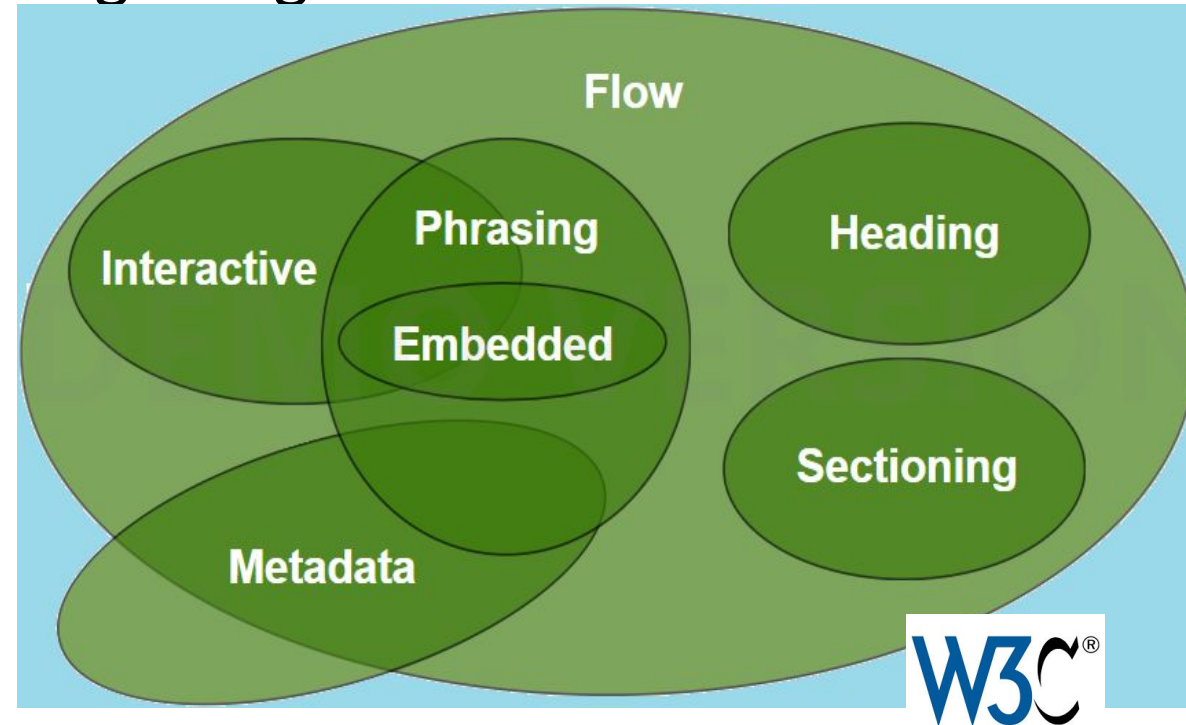
`<span>`

when and why span needed?

- when an element needs to be referred, e.g. by CSS rules or JavaScript

HTML5 Content model categories

- The categories overlap, one element may belong to several categories
- The elements are divided into the following categories
 - **flow content**: elements used in the body
 - **phrasing content**: ~ text level elements (inline elements)
 - **heading and sectioning content**: ~ block level elements
 - **embedded content**: another resource imported
 - **interactive content**: intended specifically for user interaction
 - **metadata content**: sets up the presentation



Content model of an element

- describes the element's categories, contexts and permitted contents
- additionally,
 - tag omission (empty element?)
 - permitted parents
 - permitted ARIA roles (accessibility)

4.6 Text-level semantics

4.6.1 The **a** element

Categories

[Flow content](#).

When the element only contains [phrasing content](#): [phrasing content](#).

[Interactive content](#).

Contexts in which this element can be used:

When the element only contains [phrasing content](#): where [phrasing content](#) is

Otherwise: where [flow content](#) is expected.

Content model:

[Transparent](#), but there must be no [interactive content](#) descendant.

Content attributes:

[Global attributes](#)

[href](#)

[target](#)

[rel](#)

[media](#)

[hreflang](#)

[type](#)

Links (1)

`<a href="address">link</a>`

- The link can refer to any published file in the internet

- absolute address

You can learn about links in Finnish

```
<a href="https://fi.wikipedia.org/wiki/Hyperlinkki">in Wikipedia</a>.
```

```
<a href="https://developer.mozilla.org/en-US/docs/URIs_and_URLs">URI ja URL</a>
```

```
<a href="http://merriam-webster.com">Merriam-Webster</a>
```

```
<a href="http://www.sis.uta.fi/~jwt/18/">Course material</a>
```

- besides http, supported transfer protocols include
ftp, mailto, ldap, file, news, gopher tai telnet
- after the protocol (and colon) comes the name of a server

```
<a
```

```
href="mailto:helpdesk@coop.com">helpdesk@coop.com</a>
```

Links (2)

```
<a href="src/page2.html">to the other page</a>
<a href="../../../contactinfo.html">contact us</a>
<!-- next one gives index.html if it exist, if not, it gives a directory listing -->
<a href="http://www.sis.uta.fi/~jwt/18/">lectures</a>
```

- link to a file on the same server
- relative (Unix-style file path) or absolute link

```
<h1 id="top">Heading of the page</h1>
<p> Lorem ipsum . . .</p>
. . .
:
<a href="#top">top of the page</a>
:
```

- within a page, to an element with the `id` attribute

```
<a href="https://fi.wikipedia.org/wiki/Hyperlinkki#Lähteet">More information</a> on
Hyperlinks in Finnish.
```

Link (3)

- Link has a state: **unvisited**, **visited** or **active**.
 - by default, each state has a different visual appearance
 - appearance can be changed using CSS pseudo-class selectors
- Attribute `target` defines where the content from the address is opened to
 - syntax***: `<a target="_blank" | "_self" | "_parent" | "_top">`
 - for example, attribute with the `target` value `"_blank"` opens the link to a new tab

```
<a href="contactinfo.html" target="_blank">Contact information</a>
```

Character Encoding

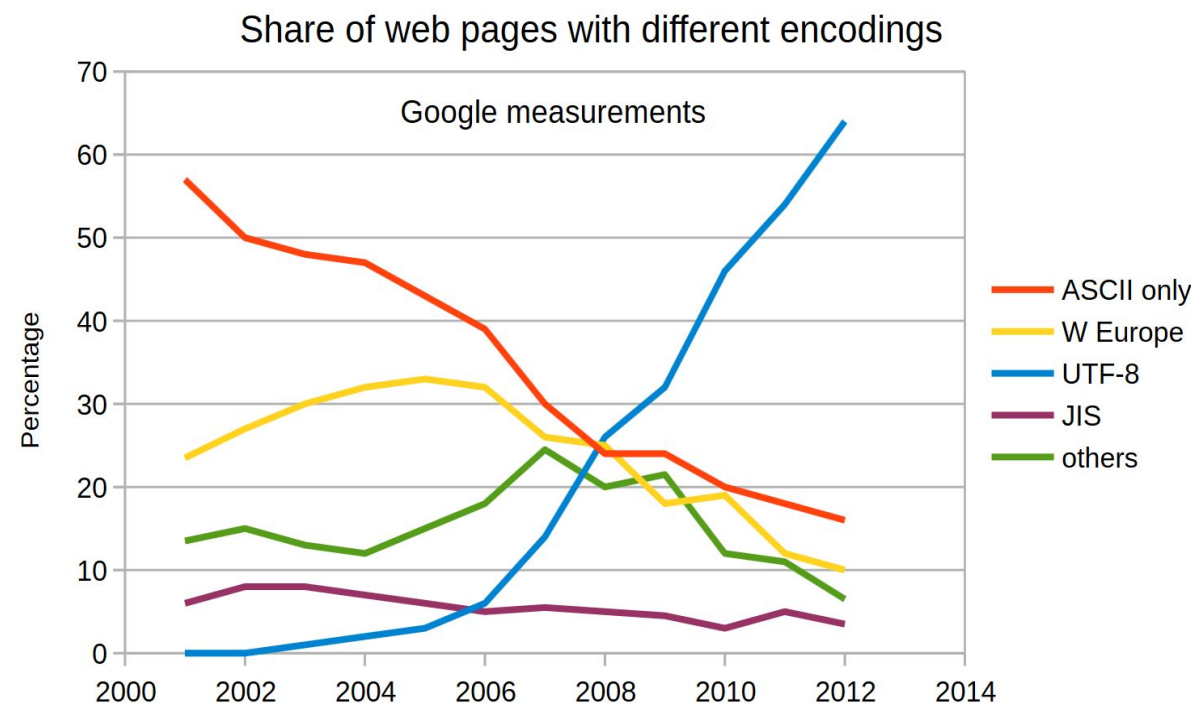
- HTML code is text saved in a file
 - done using encoding (turned into binary numbers), e.g.
 - ASCII*: Original, limited to letters, characters, some symbols, 127 characters
 - ISO/IEC-8859-1(Latin1)*: Western characters based on the Arabic script
 - ISO/IEC -8859-15*: Modified ISO-8859-1 to support Finnish, French and Estonian
 - Windows-1252*: Superset of ISO-8859-1 (additional symbols)
 - Unicode*: contains characters for most living languages and scripts in the world

still very relevant

The Absolute
Minimum Every
Software Developer
Absolutely, Positively
Must Know About
Unicode and
Character Sets (No
Excuses!)

Unicode, charset as UTF-8

- Unicode can be implemented using different encodings
- Used nowadays on web pages: UTF-8
 - uses variable length to save characters and symbols
 - capable of encoding over 1 million characters
 - backward-compatible with ASCII
 - the first 127 characters encoded with the same codes as in ASCII



Character escapes (HTML entities)

- A way of representing characters when, e.g.:
 - the symbol is not available
 - a character conflicts with the syntax (e.g., <,>)
 - a character is invisible or ambiguous
 - information is transferred using ASCII encoding (e.g. URL)
- Escapes provide a way of representing entities using ASCII only:

`&entity-name;` or `&#entity-number;`

<	→	<	also <
>	→	>	also >
&	→	&	also &
"	→	"	also "
'	→	'	also '
no-break space	→	 	also <
ä	→	ä	also ä
Ä	→	Ä	also Ä
ö	→	ö	also Ö
Ö	→	Ö	also ö